

Lecture #8: Sequential Games

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Math 190: Mathematics of Sustainability, Emory University

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EMORY
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1 Sequential Games

2 Multistage Games

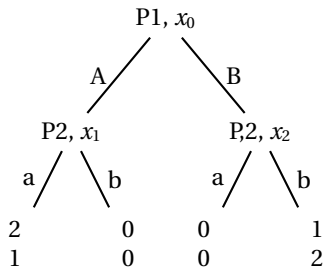
Sequential Hanging with Friends

- ① Players: $N = 2$
- ② Strategies:
 - $s_1 = \{A, B\}$
 - $s_2 = \{aa, ab, ba, bb\}$
- ③ payoffs:

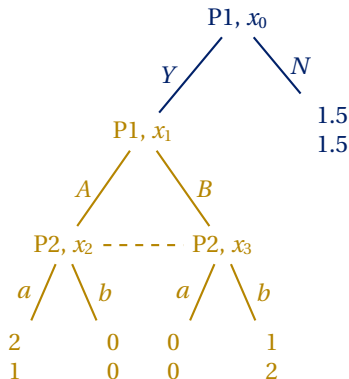
		Player 2			
		<i>aa</i>	<i>ab</i>	<i>ba</i>	<i>bb</i>
Player 1	<i>A</i>	2,1	2,1	0,0	0,0
	<i>B</i>	0,0	1,2	0,0	1,2



Not all Nash Equilibria are Subgame Nash Equilibria



Games of Imperfect Information - voluntary Hanging with Friends



Analyzing Voluntary Hanging with Friends

Preliminaries

Definition

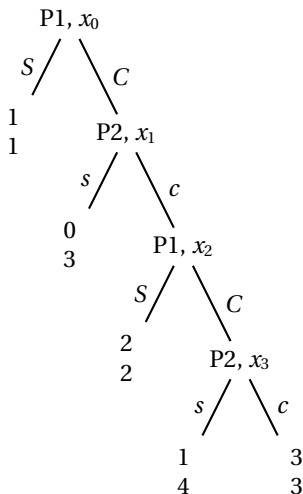
A *subgame* is a

Remark

The solution concept for finding Nash Equilibria for (smaller) subgames first and extending these to a larger game, is known as “subgame-perfect” Nash equilibria. This won someone the Nobel Prize in 1994, same year as John Nash.



Your Turn - Centipede Game



Check your Work

We start our backward strategy to see if we can find any subgame- perfect Nash Equilibria.

- If we are at x_3 , the best response is to stop since $4 > 3$.



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- if we are at x_2 , the best response is to stop since we know that we will get 1 if they choose to continue C , due to common knowledge and rationality. And they get 2 if they stop so $2 > 1$.



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- if we are at x_1 the best response is to stop because if we continue c , then they payoff will be 2.



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- if we are at x_2 , the best response is to stop since we know that we will get 1 if they choose to continue C , due to common knowledge and rationality. And they get 2 if they stop so $2 > 1$.
- if we are at x_1 the best response is to stop because if we continue c , then they payoff will be 2.
- if we are at x_0 the best response is to stop because $1 > 0$.



Cuban Missile Crisis

VOL. CXIII...No. 38,258. © 1962 by The New York Times Company. Times Square, New York 20, N. Y. NEW YORK, TUESDAY, OCTOBER 23, 1962. 13 cents (except 10-cents zone from New York City) except on late Island, higher in air delivery district. FIVE CENTS

U.S. IMPOSES ARMS BLOCKADE ON CUBA ON FINDING OFFENSIVE-MISSILE SITES; KENNEDY READY FOR SOVIET SHOWDOWN

U. S. JUDGES GIVEN POWER TO REQUIRE VOTE FOR NEGROES

High Court Upholds Order
Forcing the Registration of
54 in Alabama County

Special to The New York Times
WASHINGTON, Oct. 22 — The Supreme Court held today that Federal judges have the power to make state registration of specific Negroes on the voting rolls.

Alabama had challenged an order by Federal District Judge Frank M. Johnson Jr. requiring the registration of 54 specific Negroes in Macon County, Ala. The order was upheld by the United States Court of Appeals.

Chinese Open New Front; Use Tanks Against Indians

Nehru Warns of Peril to Independence
—Reds Attack Near Burmese Border
and Press Two Other Drives

Special to The New York Times
NEW DELHI, Oct. 22—Prime Minister Jawaharlal Nehru today told the people of India tonight that the Chinese Communist attack for us to realize fully this was a threat to their liberty. His grave warning followed news that the advancing Chinese had opened a third front in the Himalayas, near the Burmese border, and had used tanks for the first time. Five more Indian posts fell to the Chinese on the third day of savage fighting.

Excerpts from Nehru's speech will be found on Page 2.

"The time has come," he said, "when we must stand up to the Chinese Communist attack for us to realize fully this was a threat to their liberty. His grave warning followed news that the advancing Chinese had opened a third front in the Himalayas, near the Burmese border, and had used tanks for the first time. Five more Indian posts fell to the Chinese on the third day of savage fighting."

Prime Minister Nehru said India would not abandon her economic development program and policy of nonalignment with international blocs, but called on the nation to switch "from the old to the new" methods of economic development to those which produce results quickly.

"We must hold on our mill-

SHIPS MUST STOP Other Action Planned If Big Rockets Are Not Dismantled

By JAMES RESTON
Special to The New York Times
WASHINGTON, Oct. 22 — President Kennedy drew the line tonight, not with Cuba, but with the Soviet Union. After almost a generation of trying to keep the "cold war" from reaching a direct confrontation between United States and Soviet power, a decision has been made to force Soviet missile bases from this hemisphere at the risk of war.

This is the official interpretation of President Kennedy's speech tonight, and the orders to American forces bear it out.



ANNOUNCES HIS ACTION: President Kennedy speaking to the nation last night

PRESIDENT GRAVE Asserts Russians Lied and Put Hemisphere in Great Danger

By ANTHONY LEWIS
Special to The New York Times
WASHINGTON, Oct. 22 — President Kennedy imposed a naval and air "quarantine" tonight on the shipment of offensive military equipment to Cuba.

In a speech of extraordinary gravity, he told the American people that the Soviet Union, contrary to promises, was building offensive missile and bomber bases in Cuba. He said the bases could handle missiles.

Text of the President's address is printed on Page 16.

Associated Press Wirephoto

Cuban Missile Crisis- History

- ① US discovers Soviet nuclear missiles in Cuba, 70 miles from Florida in US.



Cuban Missile Crisis- History

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Cuban Missile Crisis- History

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- 2 US escalates the situation bu blockading Cuba.
- 3 The USSR then backed down, agreeing to remove its missiles from Cuba
- 4 This suggests that the United States had a credible threat along the lines of “if you don’t back off we both pay dearly.”



Cuban Missile Crisis - As a mathematical game

- Two players: Player 1 (United States) and Player 2 the USSR



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- Player 1 can choose to either ignore the incident (I), resulting in maintenance of the status quo with payoffs $(0, 0)$, or escalate the situation (E).



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- Following escalation by player 1, Player 2 can back down (B), causing it to lose face and resulting in payoffs of $(10, -10)$, or it can choose to proceed to a nuclear confrontation (N).



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- Upon this choice, the players play a simultaneous-move game in which they can either retreat (R for player 1, r for player 2) or choose Doomsday (D for player 1, d for player 2), in which the world is all but destroyed.



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 - If both call things off and retreat then they suffer a small loss due to the mobilization process and payoffs are $(-5, -5)$,

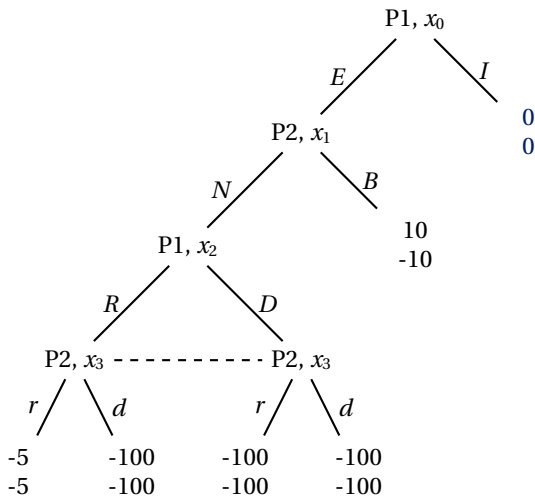


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 - If both call things off and retreat then they suffer a small loss due to the mobilization process and payoffs are $(-5, -5)$,
 - If either party chooses Doomsday then the world destructs and payoffs are $(-100, -100)$.



Cuban Missile Crisis- Decision Tree

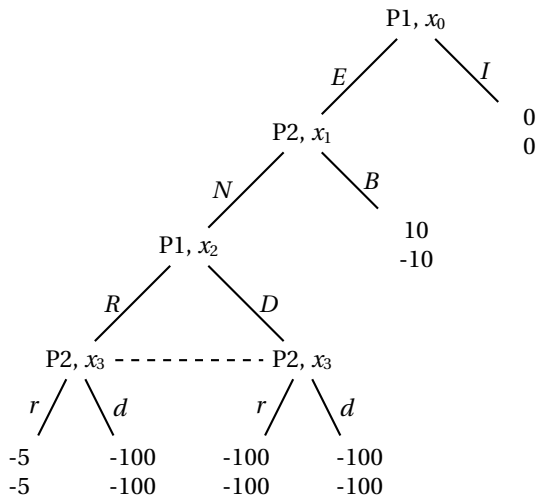


Pure Strategy Nash Equilibrium

		Player 2			
		<i>Br</i>	<i>Bd</i>	<i>Nr</i>	<i>Nd</i>
Player 1	<i>IR</i>	0,0	0,0	0,0	0,0
	<i>ID</i>	0,0	0,0	0,0	0,0
	<i>ER</i>	10,-10	10,-10	-5,-5	-100,-100
	<i>ED</i>	10,-10	10,-10	-100,-100	-100,-100



Subgame Nash Equilibrium



Subgame 1

Now to find subgame perfect equilibria

		Player 2	
		r	d
Player 1	R	-5,-5	-100,-100
	D	-100,-100	-100,-100



Subgame 2

Now we look at the subgame rooted at x_1 .

Case #1: Start with NE 1:

Case #2: Start with NE 2:

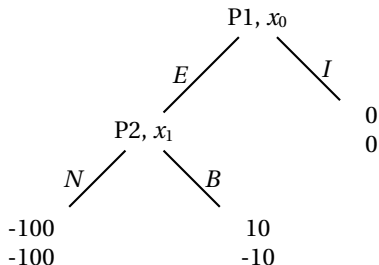
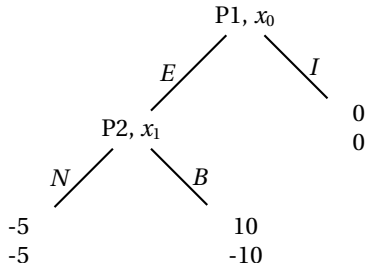


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① Sequential Games

② Multistage Games



What Are Multistage games

We define a multistage game as a finite sequence of normal-form stage-games, in which each stage-game is an independent, well-defined game of complete but imperfect information (a simultaneous-move game).



Prisoner Revenge Game

Suppose that in a first period, two players from different local neighborhoods play the familiar Prisoner's Dilemma with pure actions confess or stay silent. After this, each player can choose whether to join his local neighborhood gang or remain a “loner.” (G or L).

- if the players remain loners then they go their separate ways, regardless of the outcome of the first game, and obtain second-period payoffs of 0 each.

		Player 2	
		c	s
Player 1	C	1,1	5,-1
	S	-1,5	4,4



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- If both join gangs, then because of the nature of neighborhood gangs they will fight each other and suffer a loss, causing each to receive a payoff of -3.

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- If both join gangs, then because of the nature of neighborhood gangs they will fight each other and suffer a loss, causing each to receive a payoff of -3.
- Finally if one player joins a gang while the other player remains a loner then he loner suffers dearly since he has no gang to defend him, thus receiving a payoff of -4, while the new gang member suffers much less and receives a payoff of -1

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		c	s
Player 1	C	1,1	5,-1
	S	-1,5	4,4



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		Player 2	
		<i>l</i>	<i>g</i>
Player 1	<i>L</i>	0,0	-4,-1
	<i>G</i>	-1,-4	-3,-3



